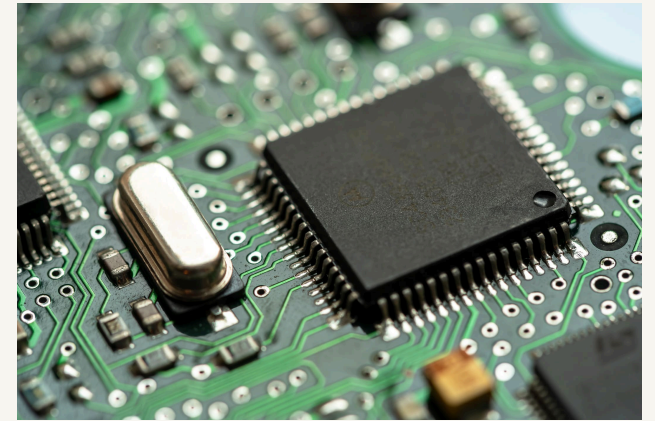
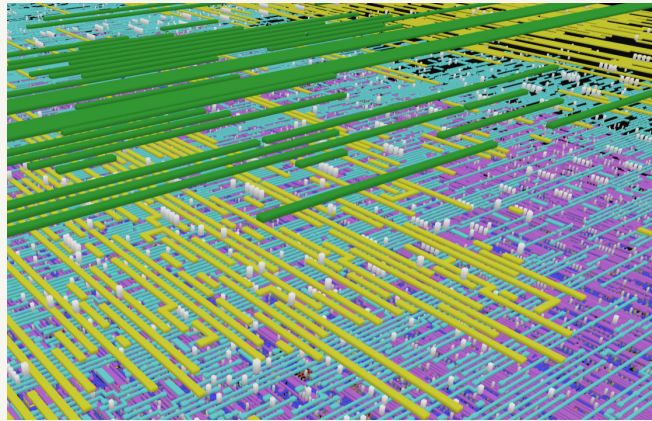


PROJECT CRATON

Ottawa is going to build a microcontroller.

Almost nobody gets to take a design all the way to the point a fab could build it.

We are doing it in public, and you could play a part in it.



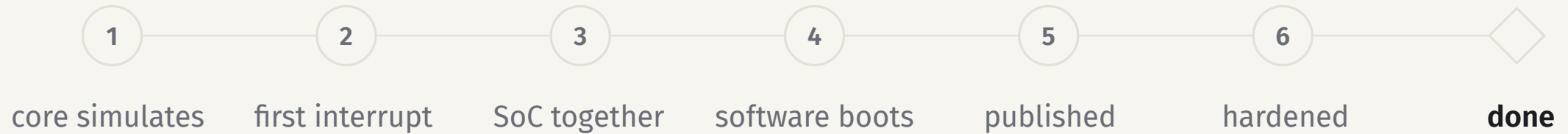
WHY THIS IS USUALLY HARD TO DO

- Universities teach the theory, not the flow.
- Industry keeps many of the final steps behind NDAs.
- The tools used to cost more than a car (maybe even a house).
- The open ones don't.

An open-source RISC-V microcontroller, built end to end with open tools (where possible). Everything we do is published as we do it.

SIX MILESTONES, AND A FINISH LINE

RISC-V Ottawa



Each one is a place somebody new could start.

The finish line is a handful of processes that anyone can run, from a clean checkout, on their own machine.

A RISC-V core simulating on your laptop, and RTL we wrote servicing real interrupts.

One container, no license, and no NDA - this is where everyone starts.

Code we wrote, running on hardware we designed, in the open so anyone can check or use it.

Integration is where we'll catch all the real bugs and edge cases, bring-up will be on a board (FPGA initially) you can hold, and code + measurement results will be published for others to look and play around with it.

The design hardened into a layout, and put through a fab's design rule check (DRC) process.

Placement, routing, timing and the physical checks, with open tools. The hardest part of the industry to get experience in. We'll also work hard to actually get it built in the end.

- A complete design taken to the point a fab could build it, this could have your name on the parts you helped build.
- SystemVerilog, assertions and coverage, synthesis, place and route, timing closure. All of it in open tools wherever possible.
- Work you can show to anyone, committed and version controlled.

The simulation and verification flow runs on any machine with a container runtime. The physical flow too (you might just need a bigger PC). Holding a contributor role will require sustained effort, not a single weekend.

THE ROLES. PICK MANY.

RISC-V Ottawa

- RTL design and verification
- RISC-V standards research
- Developing build systems usable by everyone
- Toolchain and compilers
- Embedded framework and OS ports
- FPGA work
- Physical design (RTL to GDSII)

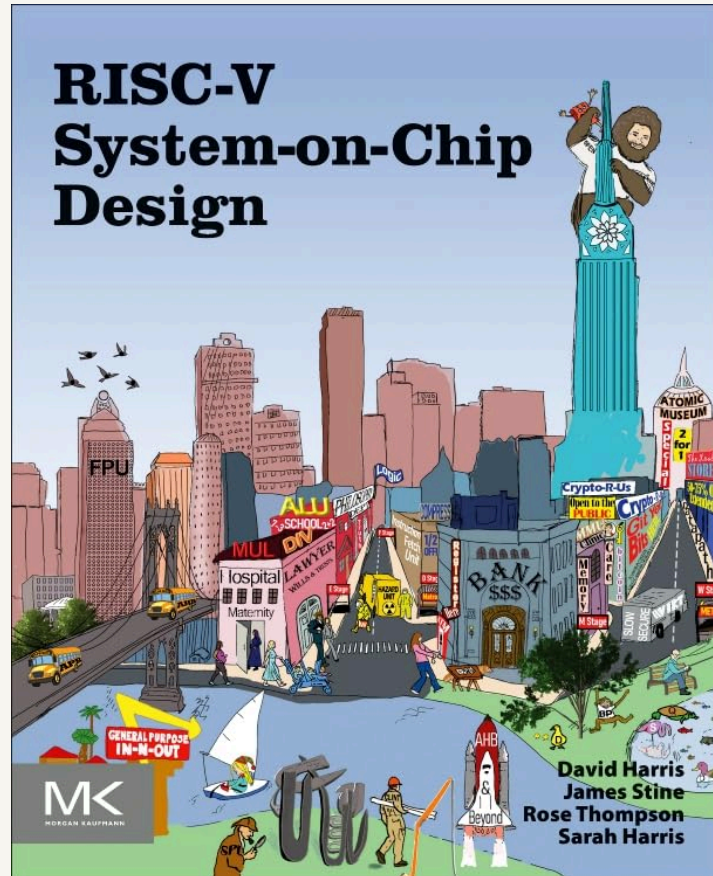
github.com/riscv-ottawa/craton

- Read the preliminary docs, fork it, provide feedback, get ready for the real work.
- License: Solderpad 2.1

The repo is barebones for now, but everything we do will be published there as we do it.

Work starts in November.

- We'll give a technical talk on what we are actually building, in November's meetup.



RISC-V System-on-Chip Design

David Harris, James Stine, Rose Thompson, Sarah Harris

- Book club!!!
- We'll read it together, a chapter at a time.
- Buy a copy and start reading with us!
- First meeting in November.

Hate books? Too expensive? Next slide!

Three doors.

Enter through at least one of them to join:

- Read the book with us.
- Join or create another RISC-V Ottawa project.
- A non-trivial PR contribution to Craton itself once it's started.

Scan the QR, fill out the short form. Whichever door you are taking, you'll hear from us in November.



Same code on the front table.